Victim Machine Setup Guide

Raspberry Pi Configuration for Recon & Password Cracking Labs

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This document is for the **instructor only**. It covers everything needed to configure the Raspberry Pi as the victim machine before the lab sessions begin. Students should not see this document.

The Pi will expose two services – SSH and FTP – and host a target user account with a deliberately weak password. Students will discover the machine via nmap, attack it online with hydra, and later extract and crack its password hash offline.

> **!** Important
>
> Complete all steps in this guide **before** students arrive. The two most time-sensitive steps are connecting the Pi to the LAN and noting its IP address, which students will need at the start of Lab 1.

1 System Preparation

1.1 Update the Pi

Start with a clean, up-to-date system:

```
sudo apt update && sudo apt upgrade -y
```

1.2 Assign a Static IP Address

The lab is designed around the Pi being reachable at a known address in the `10.7.7.0/24` subnet. A static IP prevents the Pi from disappearing between lab sessions if the DHCP lease changes.

Current Raspberry Pi OS (Bookworm and later) manages networking with **NetworkManager**, not `dhcpcd`. The `/etc/dhcpcd.conf` file no longer exists on a fresh image, so configure the static address with `nmcli`.

First, find the name of the wired connection:

```
nmcli device status          # note the DEVICE (usually eth0)
nmcli connection show        # note the connection NAME (usually "Wired
 ↪  connection 1")
```

Set the static address, gateway, and DNS on that connection, then bring it back up to apply:

```
sudo nmcli connection modify "Wired connection 1" \
    ipv4.method manual \
    ipv4.addresses 10.7.7.200/24 \
    ipv4.gateway 10.7.7.7 \
    ipv4.dns 10.7.7.7
sudo nmcli connection up "Wired connection 1"
ip addr show eth0   # confirm the address is now 10.7.7.200
```

The setting is stored under `/etc/NetworkManager/system-connections/` and persists across reboots. If you prefer a menu-driven interface, `sudo nmtui` does the same thing interactively.

> **Older Pi OS images (Bullseye and earlier)**
>
> Images from before Bookworm (released October 2023) still use `dhcpcd`. On those, edit `/etc/dhcpcd.conf` instead and append:
>
> ```
> interface eth0
> static ip_address=10.7.7.200/24
> static routers=10.7.7.7
> static domain_name_servers=10.7.7.7
> ```

then `sudo systemctl restart dhcpcd`.

> **Note**
>
> `10.7.7.200` is used throughout both student lab documents as the Pi's address. If you use a different address, do a find-and-replace in both QMD files before printing.

2 Setting Up User Accounts

The Pi hosts four user accounts in total. **`labuser`** is the primary target for the online attack in Lab 1 and is also a sudoer, allowing students to escalate to root at the start of Lab 2. The remaining three accounts – **`alice`**, **`bob`**, and **`carol`** – exist only to populate the shadow file with hashes of varying difficulty for the offline cracking exercises.

The table below summarises all four accounts. Keep this page secure – do not share it with students.

User	Password	Cracking method	Notes
`labuser`	`sunshine1`	Hydra (Lab 1), then confirms setup in Lab 2	Must be in rockyou; used for SSH/FTP
`alice`	`dragon`	Dictionary (rockyou, no rules)	Near the top of rockyou – cracks in seconds
`bob`	`c0bra`	Rule-based (`best64`)	Leet form not in rockyou; best64's `o->0` rule derives it from `cobra`
`carol`	`abc123`	Dictionary (rockyou, no rules)	Mid-list rockyou entry – takes a moment

2.1 Create labuser and Grant Sudo Access

```
sudo adduser labuser
# enter 'sunshine1' (or your chosen rockyou password) when prompted
```

Add `labuser` to the `sudo` group so students can escalate to root in Lab 2:

```
sudo usermod -aG sudo labuser
```

Verify the group membership:

```
groups labuser
# should include 'sudo' in the output
```

Test that escalation works:

```
su - labuser
sudo -i
# should drop into a root shell
exit
exit
```

> **Note**
>
> `labuser` having sudo rights is what allows the `sudo -i` command in the offline lab to succeed. Without this step, students will be denied when they try to read `/etc/shadow`.

2.2 Create the Practice Accounts

These accounts do not need sudo access – they exist only to add entries to the shadow file.

```
# alice -- plain dictionary password
sudo adduser --disabled-login alice
echo "alice:dragon" | sudo chpasswd

# bob -- requires rule-based attack to crack
sudo adduser --disabled-login bob
echo "bob:c0bra" | sudo chpasswd

# carol -- plain dictionary password
sudo adduser --disabled-login carol
echo "carol:abc123" | sudo chpasswd
```

The `--disabled-login` flag prevents these accounts from being used to log in interactively via SSH, so students cannot simply try to connect as alice, bob, or carol – they must crack the hashes offline.

2.3 Verify All Shadow Entries

```
sudo grep -E "labuser|alice|bob|carol" /etc/shadow
```

You should see four lines, each with a hash beginning with `$6$` (or `$y$` on newer Pi OS images – see note below). Confirm no entry has `*` or `!` in the hash field, which would indicate no password is set.

> **Note**
>
> If any hash begins with `$y$` rather than `$6$`, the system is using yescrypt (hashcat mode `400`) instead of sha512crypt (mode `1800`). Either update both student lab documents to use `-m 400`, or force SHA-512 by setting the password hashing algorithm explicitly:
>
> ```
> sudo sed -i 's/^ENCRYPT_METHOD.*/ENCRYPT_METHOD SHA512/' /etc/login.defs
> ```
>
> Then recreate the affected passwords with `chpasswd` as above.

2.4 Verify labuser's Password is in rockyou

```
grep -n "^sunshine1$" /usr/share/wordlists/rockyou.txt
```

This should return a line number. If it returns nothing, choose a different password from the suggestions below and update `labuser` accordingly:

```
sunshine1   dragon99   password1   princess   monkey123
```

Alice's and carol's passwords (`dragon` and `abc123`) are confirmed entries in rockyou and do not need checking. Bob's password (`c0bra`) is *intentionally* absent from rockyou in that exact form – its leetspeak spelling (`o->0`) does not appear in the list, which is what forces a rule-based attack. The base word `cobra` *is* in rockyou, and that is what lets `best64` derive `c0bra` from it.

3 Installing and Configuring SSH

SSH is almost certainly already installed and running on a Raspberry Pi OS image. Verify it:

```
sudo systemctl status ssh
```

If it is not running, install and enable it:

```
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
```

Confirm it is listening:

```
sudo ss -tlnp | grep :22
```

> **Tip**
>
> `ss` (from the `iproute2` package) is the modern replacement for the deprecated `netstat`; it is preinstalled on current Pi OS. If you still prefer `netstat`, install it with `sudo apt install net-tools -y`.

3.1 Allow Password Authentication

Some Pi OS images disable password authentication by default in favour of key-based login. Hydra needs password authentication to be enabled.

```
sudo nano /etc/ssh/sshd_config
```

Ensure the following lines are present and uncommented:

```
PasswordAuthentication yes
PermitRootLogin yes
```

Setting `PermitRootLogin yes` is necessary for students to SSH in as root at the start of Lab 2 after cracking `labuser`'s password and escalating. Restart SSH to apply:

```
sudo systemctl restart ssh
```

> **Important**
>
> `PermitRootLogin yes` is a significant security risk on any real system. This is acceptable here because the Pi is on an isolated LAN. Revert this setting when the lab is over.

4 Installing and Configuring FTP

Students already know vsftpd from the covert channels lab. Having a second service on the Pi makes the nmap output more interesting and gives hydra two protocols to attack.

4.1 Install vsftpd

```
sudo apt install vsftpd -y
```

4.2 Configure vsftpd

```
sudo nano /etc/vsftpd.conf
```

Find and set the following options:

```
anonymous_enable=NO
local_enable=YES
write_enable=NO
local_umask=022
chroot_local_user=YES
allow_writeable_chroot=YES
```

Setting `local_enable=YES` allows system users (including `labuser`) to log in with their system credentials. This is what hydra will attack.

Restart vsftpd:

```
sudo systemctl enable vsftpd
sudo systemctl restart vsftpd
```

Verify it is listening on port 21:

```
sudo ss -tlnp | grep :21
```

5 Final Verification

Before students arrive, run through this checklist:

1. Pi is connected to the LAN and reachable at `10.7.7.200`:

```
ping 10.7.7.200   # from another machine on the LAN
```

2. SSH is open on port 22 and accepts password login:

```
ssh labuser@10.7.7.200   # should prompt for password
```

3. FTP is open on port 21 and accepts `labuser` credentials:

```
ftp 10.7.7.200   # login as labuser
```

4. `labuser` can escalate to root:

```
ssh labuser@10.7.7.200
sudo -i           # should drop to a root shell without errors
exit
exit
```

5. All four shadow entries are present and hashed (not `*` or `!`):

```
sudo grep -E "labuser|alice|bob|carol" /etc/shadow
```

All four lines should have a `$6$` (or `$y$`) hash in the second field.

6. The `labuser` password appears in rockyou.txt:

```
grep -n "^sunshine1$" /usr/share/wordlists/rockyou.txt
```

This should return a line number. If it returns nothing, the password will not be crackable – choose a different one from the suggestions in the user accounts section above.

7. The practice account passwords are confirmed crackable:

```
grep -n "^dragon$"   /usr/share/wordlists/rockyou.txt   # alice
grep -n "^abc123$"   /usr/share/wordlists/rockyou.txt   # carol
```

For `bob`, the rule-based attack depends on two things being true: the leet form `c0bra` must be *absent* from rockyou, while the base word `cobra` must be *present* so `best64` can derive it:

```
grep -n "^c0bra$"   /usr/share/wordlists/rockyou.txt   # bob -- should return
↪  NOTHING
grep -n "^cobra$"   /usr/share/wordlists/rockyou.txt   # should return a line
↪  number
```

If `c0bra` ever turns up in your copy of rockyou, pick another leet word whose plain form is in the list but whose substituted form is not (e.g. `v1per` from `viper`) and update `bob` accordingly.

Once all seven checks pass, the Pi is ready.

6 Teardown After the Lab

When the lab sessions are over, revert the security-sensitive changes:

```
# Disable root SSH login
sudo nano /etc/ssh/sshd_config
# Set: PermitRootLogin no
sudo systemctl restart ssh

# Remove all lab user accounts
sudo deluser --remove-home labuser
sudo deluser --remove-home alice
sudo deluser --remove-home bob
sudo deluser --remove-home carol

# Optionally stop vsftpd if not needed outside the lab
sudo systemctl disable vsftpd
sudo systemctl stop vsftpd
```